

ASSIGNMENT ON CSE 422

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Section : 11

Assignment No: 2

Answer to the Question No.1

(A)

Genetic Algo are iterative optimization technique which produces natural selection to find optimal solutions.

Elitism is the concept of prioritize one over another on a certain metric.

During generation update in the algorithm it chooses which old specimen's should be carried on. It guarantees the survival of the best solution found so far. Thus, it prevents the algorithm from losing a good solution due to nature of selection, cross over or mutation.

(B)

Cross over is the method used to combine parents and offspring.

→ Single point: A single point is selected randomly along the length.

Then the two segments are swapped.

$$\begin{array}{cc|c} 1101 & 0110 & \text{--- i) } 10010110 \\ 1001 & 1011 & \text{--- ii) } 11011011 \end{array}$$

Two point Crossover : Two points are randomly selected at length. The genetic material between these 2 segments are swapped.

$$\begin{array}{c|c|c} 11 & 01 & 10 \\ \hline 00 & 11 & 01 \end{array}$$

(i) 11 11 010

ii 00 01 01

So, these are the single and two point cross-over.

(c)

Some stopping conditions in Genetic Algo implement :-

(i) Maximum generations :

For a fixed no. of generations the algorithm terminates

(ii) Fitness Metric

(iii) No improvement is found.

(b)

When we randomly alter one or more gene in a chromosome we term it as mutation. The importance of mutation is:-

(i) It introduces new genetic variation into the population

(ii) Mutation prevents the population from converging too early on a sub optimal solution

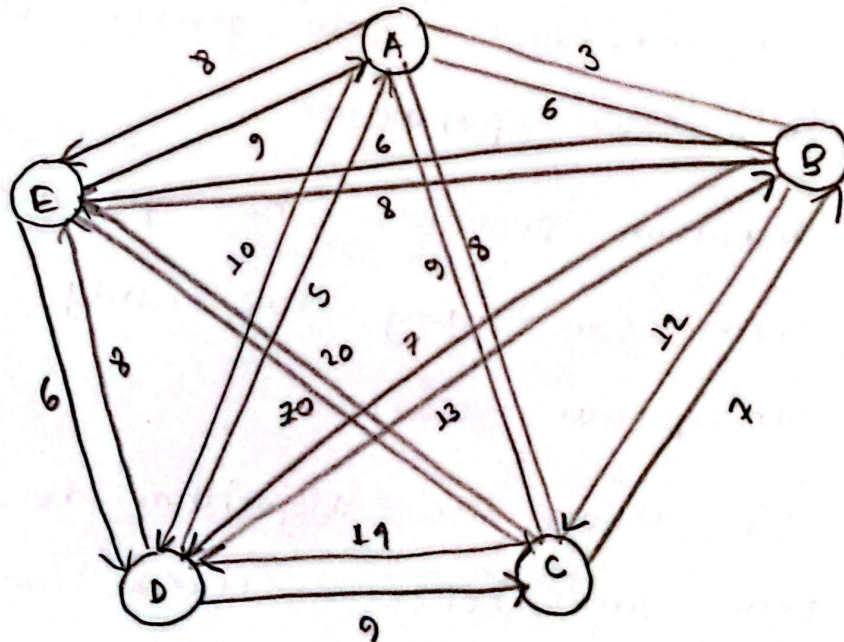
(iii) It allows the algorithm to explore new and better solutions.

(E)

Crossover is the action of combining genetic info from two parent individuals to create an offspring.

- (i) Effective search of the solution space
- (ii) Combines the successful traits of the fittest ones.

Answer to the Question No. 2



(a)

Chromosome Representation

$c_1: C, E, A, B, D$

$c_2: E, D, A, C, B$

$c_3: D, C, B, A, E$

$c_4: A, B, C, D, E$

Fitness Calculation

As, it is required to find the lowest cost so this is a minimization problem.

$$\text{So, Fitness} = \frac{1}{\text{Distance}}$$

$$\begin{aligned} c_1: & C \xrightarrow{9} E \xrightarrow{6} A \xrightarrow{8} B \xrightarrow{12} D \xrightarrow{7} C \\ & = 42 \end{aligned}$$

$$\begin{aligned} c_2: & E \xrightarrow{11} D \xrightarrow{9} A \xrightarrow{8} C \xrightarrow{6} B \xrightarrow{9} E \\ & = 46 \end{aligned}$$

$$\begin{aligned} c_3: & D \xrightarrow{7} C \xrightarrow{6} B \xrightarrow{8} A \xrightarrow{6} E \xrightarrow{14} D \\ & = 41 \end{aligned}$$

$$C4: A \xrightarrow{8} B \xrightarrow{6} C \xrightarrow{7} D \xrightarrow{14} E \xrightarrow{6} \dots \rightarrow A$$

$$= 41$$

So, the fitness values will be: -

$$C1 \rightarrow 1/42 = 0.0238$$

$$C2 \rightarrow 1/46 = 0.0217$$

$$C3 \rightarrow 1/41 = 0.02439$$

$$C4 \rightarrow 1/41 = 0.02439$$

(b)

We are asked to select two best chromosomes using Roulette wheel or Rank Based

Roulette Wheel Selection:

$$\text{sum of the fitness} \rightarrow 0.0943$$

$$\begin{array}{r} C1 \rightarrow 0.2620 \\ \hline 0.0943 \end{array}$$

$$\begin{array}{r} C2 \\ \hline 0.0943 \end{array} \rightarrow 0.23011$$

$$C3/0.0943 \rightarrow 0.2587$$

$$C4/0.0943 \rightarrow 0.2587$$

Now, we select two chromosomes based on these probabilities. We choose C_3 and C_4 with 0.2687

(c)

So, we apply single point or uniform cross over.

Parent 1: A B C D E A

Parent 2: D C B A E D

A B C | D E A

D C B | A E D

→ A B C A E D

→ D C B D E A

So, the 2nd phase,

A B | C D E A

D C | B A E D

→ A B B A E D

→ D C C D E A

So, 4 children that are generated is:-

A B C A E D, D C B D E A, A B C D E D, D C B A E A

(d)

So, we perform single bit mutation:

$C'_1 \rightarrow A B C A E D \longrightarrow A E C A B D [C''_1]$

$C'_2 \rightarrow D C B D E A \longrightarrow D E B D C A [C''_2]$

$C'_3 \rightarrow A B C D E D \longrightarrow A E C D B D [C''_3]$

$C'_4 \rightarrow D C B A E A \longrightarrow D E B A C A [C''_4]$

We interchange the 2nd & 5th position.

A valid chromosome in TSP should contain all the cities exactly one. So,

$C''_1 \rightarrow$ Not valid

$C''_2 \rightarrow$ Not valid

$C''_3 \rightarrow$ Not valid

$C''_4 \rightarrow$ Not valid

Since we see all children's chromosome are not valid, we can consider the process of multi-bit mutation.

$C''_1 \rightarrow A$ duplicate

$C''_2 \rightarrow D$ Duplicate

$C''_3 \rightarrow D$ Duplicate

$C''_4 \rightarrow A$ Duplicate

So, even after applying multi-bit mutation we are unable to solve the problem.

Answer to the Question No. 3

(a)

Our assumption of chromosomes in this 8-queen problem is:-

$C_1 : 8 \ 6 \ 4 \ 2 \ 7 \ 5 \ 3 \ 1$

$C_2 : 2 \ 4 \ 7 \ 5 \ 8 \ 1 \ 3 \ 6$

$C_3 : 4 \ 1 \ 5 \ 8 \ 2 \ 7 \ 3 \ 6$

$C_4 : 5 \ 3 \ 1 \ 6 \ 8 \ 2 \ 4 \ 7$

The fitness function for this problem is
↳ Avoiding clashes.

Thus → no. of non attacking queen pairs

Maximum fitness for 8 Queen problem
would be:- $8C_2 = 28$

$C_1 \rightarrow$ Attacking pair : 1

$C_2 \rightarrow$ " : 3

$C_3 \rightarrow$ " : 1

$C_4 \rightarrow$ " : 2

So, fitness for C_1 , C_2 , C_3 and C_4
will be 27, 25, 27, 26

(b)

We are supposed to pick 2 chromosomes
for natural selection. So,

Parent 1 (C_1): [8, 6, 4, 2, 7, 5, 3, 1]

Parent 2 (C_3): [4, 1, 5, 8, 2, 7, 3, 6]

(c)

We are supposed to create 4 children using single point cross-over method.

8	6	4	2		7	5	3	1
4	1	5	8		2	7	3	6

→ 8 6 4 2 2 7 3 6

→ 4 1 5 8 7 5 3 1

Now we treat other two offsprings

8	6	4	2	7	5		3	1
4	1	5	8	2	7		3	6

→ 8 6 4 2 7 5 3 6

→ 4 1 5 8 2 7 3 1

(d)

Now, we are required to perform mutation on the offsprings

$O_1 \rightarrow$	8	6	4	2	2	7	3	6
$O_2 \rightarrow$	4	1	5	8	7	5	3	1
$O_3 \rightarrow$	8	6	4	2	7	5	3	6
$O_4 \rightarrow$	4	1	5	8	2	7	3	1

} Mutation
(changing the 2nd gene with $(8-x)$)

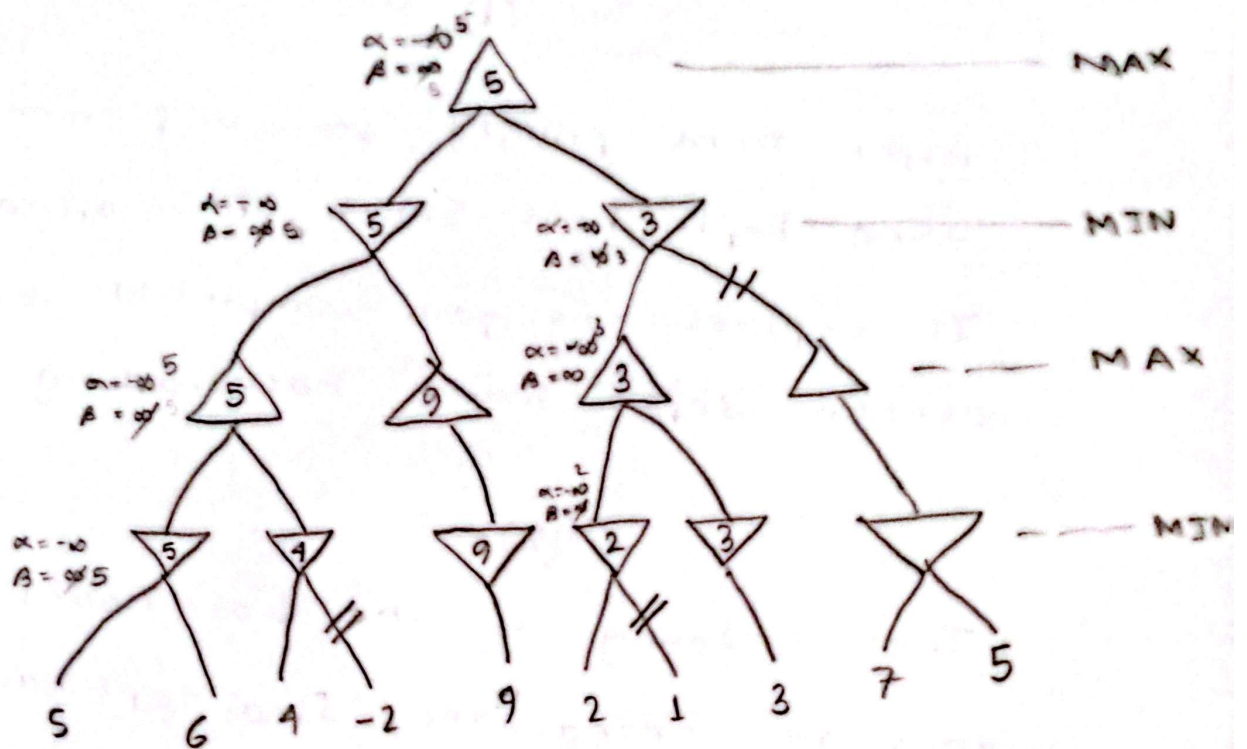
$O'_1 \rightarrow$ 8 2 4 2 2 7 3 6
 $O'_2 \rightarrow$ 4 7 5 8 7 5 3 1
 $O'_3 \rightarrow$ 8 2 4 2 7 5 3 6
 $O'_4 \rightarrow$ 4 7 5 8 2 7 3 1

(e)

The calculations for the offspring are:-

$O'_1 \rightarrow 28 - 6 = 22$
 $O'_2 \rightarrow 28 - 5 = 23$
 $O'_3 \rightarrow 28 - 2 = 26$
 $O'_4 \rightarrow 28 - 2 = 26$

Answer to the Question No. 4



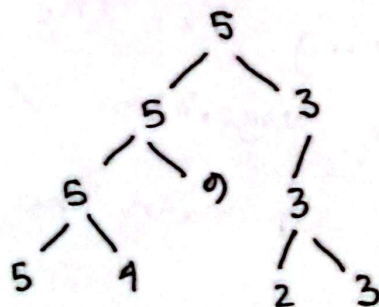
We know,

$$\alpha = -\infty, \beta = +\infty$$

Alpha Beta Pruning,

$$\alpha \geq \beta$$

So, the root value will be 5



Answer to the Question No 5

(1)

Alpha-Beta pruning is implemented using Depth First Search (DFS) approach. It explores one path completely to a certain depth before backtracking.

(2)

The ordering of the leaf nodes does not change the final optimal value which is determined using min-max algorithm.

(3)

Yes we can obtain the same results using min-max and Alpha-Beta pruning. Alpha-Beta pruning is the optimization of the min-max algorithm. So, both could essentially end up with the same optimal move.